

Olist E-Commerce SQL Analysis

ONE-PAGE INSIGHT SUMMARY

1. DATA OVERVIEW

This analysis examines the Olist Brazilian e-commerce dataset across **8 related tables**. The orders data contains **99,441 orders** from **September 4, 2016 to October 17, 2018**, while 99,441 customer records correspond to **96,096 unique people**. Data-quality checks found missing order lifecycle dates, **610 products without a category**, and monetary fields stored as text; monetary values were converted to numeric form for analysis.

2. THREE BUSINESS INSIGHTS

Repeat purchasing is very limited (Q7).

Of 96,096 unique customers, **93,099 (96.88%)** are one-time buyers and only **2,997 (3.12%)** are repeat buyers. This indicates that the marketplace depends heavily on customers who purchase only once, making second-purchase conversion a major opportunity.

High-spend customers have substantially greater value (Q6).

The three spending tiers contain roughly one-third of customers each, but average spend rises from **50.00** in the Low tier to **109.99** in Medium and **339.80** in High. This suggests that customer value differs sharply even when the customer base is divided into equal-sized groups.

The leading category is important, but revenue remains diversified (Q4 & Q8).

beleza_saude (Beauty & Health) generates the most categorized product revenue at **1,258,681.34**. However, it represents only **9.38%** of the **13,412,108.42** categorized product revenue measured in Q8, so overall product revenue is not concentrated in one category.

3. TWO RECOMMENDATIONS

1. Build a focused second-purchase strategy.

Prioritize retention initiatives for first-time buyers, such as post-purchase follow-ups, personalized offers, and incentives designed to encourage a second order. Track repeat-purchase rate over time to test whether these actions improve the current 3.12% repeat-buyer share.

2. Target high-value customers while keeping category investment diversified.

Use spending segments to identify and engage customers with the greatest value potential, since the High tier averages 339.80 in spend. At the same time, support strong categories such as Beauty & Health without overconcentrating resources, because the top category contributes less than one-tenth of categorized product revenue.

Scope note

Customer/payment analyses use payment value; product/category analyses use item price. Q8's percentage is therefore the share of categorized product revenue.